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INTRODUCTION TO NETWORKING WITH HACK THE BOX

1.1 Introduction

In this module , I have learnt the fundamentals of networking including the network topology that really stood out to me. How you place the host and the client really matter. This module has been really broad , it also brought back how to calculate ipv4 with binary numbers, subnet masking and the famous OSI model. It's one of those modules where everything feels theoretical until you realize that every time you load a webpage, your computer is doing all that math and following those OSI layers in milliseconds.

1.2 Subnetting.

Subnetting is dividing the IPV4 into small addresses.

Question.

a) Submit the decimal representation of the subnet mask from the following CIDR:
10.200.20.0/27

Answer: **225.255.255.224**

Explanation.

The /27 is the CIDR notation showing the bits that are used in the network.

First 27 bits = **1** (network bits)

Last 5 bits = **0** (host bits) because IPv4 has 32 bits, divided into 4 octets (8 bits each).

Meaning the first 3 octets:

are all 1 of 8 = 11111111, 11111111, 11111111, 11100000.

To calculate this in binary:

$128 + 64 + 32 + 24 + 8 + 4 + 2 + 1 = 255$ for the first 3 octets.

For the fourth one , i added only the 1's:

$128 + 64 + 32 = 224$ so that i got the whole subnet mask to be = 255.255.255.224

Questions

Answer the question(s) below to complete this Section and earn cubes!

+ 2 🟢 Submit the decimal representation of the subnet mask from the following CIDR: 10.200.20.0/27

255.255.255.224

Submit

b) Submit the broadcast address of the following CIDR: 10.200.20.0/27

Answer: 10.200.20.31

Explanation:

Since $27 = 27, 1$'s and $5, 0$'s this is the last octet of the IP address.

1110000

When calculating the broadcast address, we set all host bits = 1 $\rightarrow$ 11111 because the last 5 bits in the 4th octet are host bits: 00000.

We use binary to calculate again.

128, 64, 32, 16, 8, 4, 2, 1

0 0 0 1 1 1 1 1

Total $-16+8+4+2+1=31$

So the broadcast = 10.200.20.31

+ 2 🟢 Submit the broadcast address of the following CIDR: 10.200.20.0/27

10.200.20.31

Submit

c) Split the network 10.200.20.0/27 into 4 subnets and submit the network address of the 3rd subnet as the answer.

Answer: **10.200.20.16**

Explanation:

The normal IP address is 10.200.20.0 /27

$32-27=5$

New host bits $:5-2=3$

host bits = 3 $\rightarrow$ total addresses = $2^3 = 8$ per subnet

So each subnet starts 8 addresses apart to arrange them as below:

List the subnets starting from 10.200.20.0

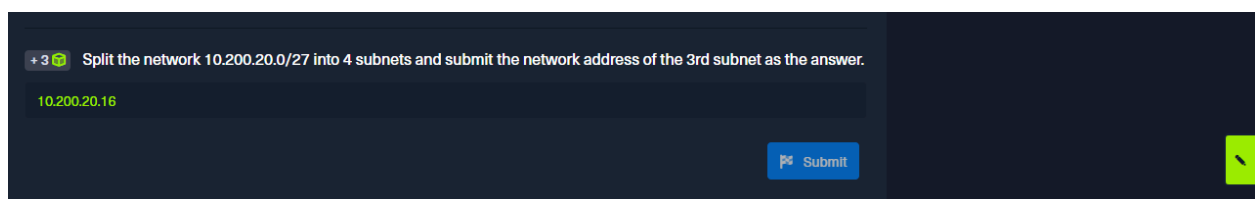
1.10.200.20.0

2.10.200.20.8

3.10.200.20.16

4.10.200.20.24

Correct answer: 10.200.20.16



A screenshot of a quiz interface. The question text is "Split the network 10.200.20.0/27 into 4 subnets and submit the network address of the 3rd subnet as the answer." Below the question, the answer "10.200.20.16" is entered in a text box. A blue "Submit" button is visible to the right of the text box. A green checkmark icon is in the top right corner of the interface.

d) Split the network 10.200.20.0/27 into 4 subnets and submit the broadcast address of the 2nd subnet as the answer.

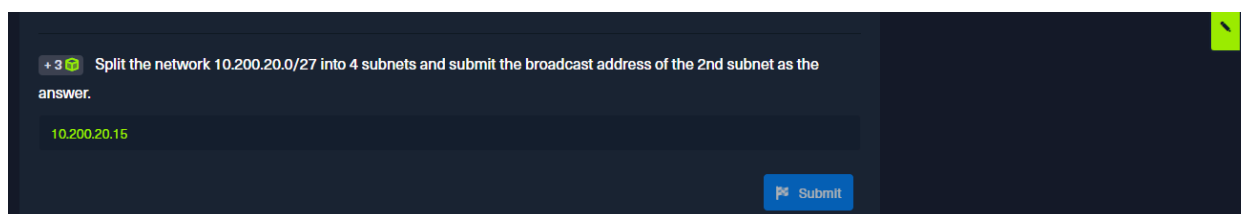
Answer: **10.200.20.15**

Explanation:

I used this list to assist me get the broadcast address, by counting the total address and adding 8 but counting from 0 .Meaning for broadcast will be an increment of 7.

List the subnets starting from 10.200.20.0	Broadcast address
1.10.200.20.0	$10.200.20.0+7= 10.200.20.7$
2.10.200.20.8	$10.200.20.8+7=10.200.20.15$
3.10.200.20.16	$10.200.20.16+7=10.200.20.23$
4.10.200.20.24	$10.200.20.24+7=10.200.20.31$

So 2nd broadcast address =10.200.20.15



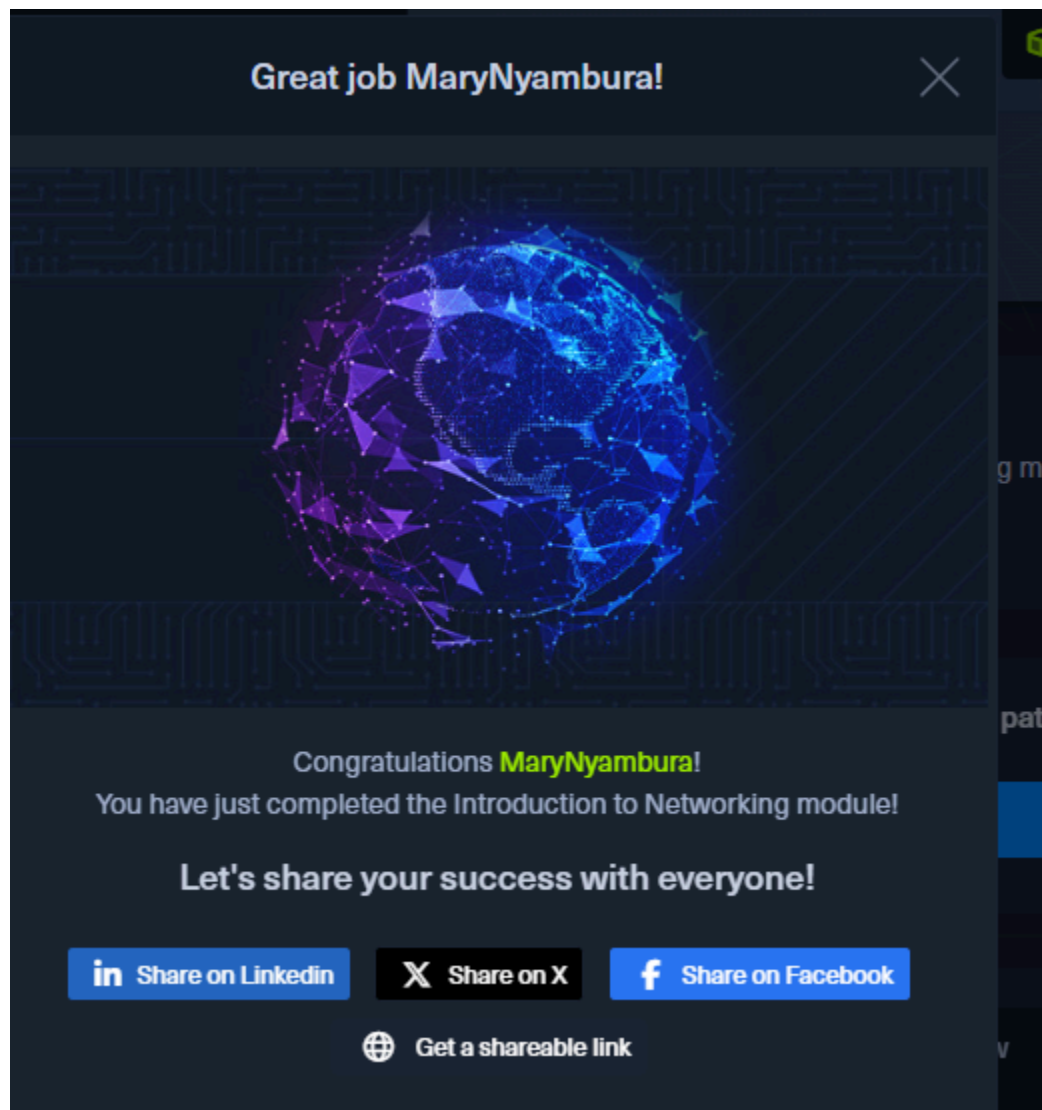
A screenshot of a quiz interface. The question text is "Split the network 10.200.20.0/27 into 4 subnets and submit the broadcast address of the 2nd subnet as the answer." Below the question, the answer "10.200.20.15" is entered in a text box. A blue "Submit" button is visible to the right of the text box. A green checkmark icon is in the top right corner of the interface.

1.3 Conclusion and Summary

As I went through the networking module, I gained a deeper understanding of how information travels across networks, how it is broken down into packets, and how these packets are routed to their destinations. I also learned about encryption and decryption using symmetric and asymmetric keys, which gave me insight into how data can be secured from unauthorized access. The subnetting part was particularly eye-opening, as it taught me the logic behind IPv4 addressing and how networks can be segmented. All of this knowledge will help me as a security analyst by enabling me to better protect networks, monitor traffic for potential threats, implement secure network architectures, and understand how attackers might try to exploit weaknesses in network design.

Below is a screenshot of my achievement completion and link.

<https://academy.hackthebox.com/achievement/2382167/34>



Great job MaryNyambura!



Congratulations **MaryNyambura!**

You have just completed the Introduction to Networking module!

Let's share your success with everyone!

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